

EASTERN WEST VIRGINIA, WV, USA

WMO: 724177

Lat: 39.402N

Lon: 77.984W

Elev: 163

StdP: 99.38

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13734

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.6	-9.2	-20.5	0.6	-9.2	-17.9	0.8	-7.5	12.4	1.6	11.3	0.6	3.1	310	0.448

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	33.6	23.2	32.1	22.8	30.8	22.2	24.9	30.8	24.1	29.6	23.4	28.6	3.8	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.1	27.2	22.4	17.5	26.6	21.9	16.9	26.0	76.1	30.8	73.0	29.6	70.3	28.6	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.5	7.5	DB	-16.3	36.2	3.3	2.0	-18.6	37.6	-20.5	38.7	-22.3	39.8	-24.7	41.3	(4)
(5)			WB	-16.9	26.4	3.1	1.1	-19.1	27.1	-20.9	27.8	-22.6	28.4	-24.8	29.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.6	0.3	1.8	6.2	12.4	17.2	22.0	24.3	23.3	19.6	13.2	7.3	2.5
	DBStd	9.49	5.89	5.32	5.28	4.72	4.29	3.23	2.74	2.63	3.69	4.44	4.67	4.83
	HDD10.0	1072	307	234	141	27	2	0	0	0	0	17	107	237
	HDD18.3	2683	560	462	376	186	74	7	0	1	29	168	331	489
	CDD10.0	2012	6	5	25	99	225	360	442	413	287	117	26	6
	CDD18.3	583	0	0	1	8	39	117	184	156	66	11	0	0
	CDH23.3	5437	0	3	15	127	406	1076	1769	1371	568	99	4	1
	CDH26.7	1964	0	0	2	32	120	375	732	497	186	21	0	0
Wind	WSAvg	2.7	3.3	3.4	3.5	3.3	2.5	2.3	2.1	1.8	2.0	2.4	2.8	2.9
Precipitation	PrecAvg	967	64	61	87	79	102	91	91	82	85	78	77	67
	PrecMax	1529	161	148	198	198	246	234	221	182	238	241	199	162
	PrecMin	669	12	4	7	4	9	17	9	14	4	2	8	13
	PrecStd	185	35	37	40	42	59	51	49	43	63	54	44	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.6	21.2	25.9	30.1	32.2	34.4	36.4	34.3	33.5	29.1	23.3	20.2
		MCWB	14.3	13.8	15.7	17.8	21.6	23.7	23.8	23.2	22.2	21.1	15.3	14.6
	2%	DB	15.8	16.6	21.6	27.0	29.9	32.2	34.0	32.5	31.1	26.2	20.2	16.0
		MCWB	12.0	10.3	13.4	16.8	20.9	23.0	23.5	23.2	21.3	18.9	14.1	11.7
	5%	DB	12.1	12.9	18.4	24.1	27.7	30.6	32.3	31.0	28.8	23.5	17.8	12.5
		MCWB	8.6	7.7	12.1	15.5	19.5	22.3	23.0	22.6	20.7	17.3	12.9	9.0
	10%	DB	8.5	9.9	15.6	21.4	25.4	28.9	30.7	29.5	26.7	21.3	15.4	9.9
		MCWB	5.3	5.7	10.1	13.9	18.5	21.3	22.5	21.9	19.7	16.2	10.7	6.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.6	15.1	17.1	19.9	23.4	25.4	25.8	25.7	24.4	22.4	17.5	15.8
		MCDB	18.1	19.7	24.3	26.6	29.7	31.3	32.8	31.5	29.3	26.6	20.7	19.0
	2%	WB	12.8	11.1	15.0	18.3	21.9	24.1	25.0	24.7	23.2	20.7	15.7	12.6
		MCDB	15.2	15.1	19.7	24.6	27.5	30.0	31.4	30.0	27.5	24.4	18.9	15.1
	5%	WB	8.7	8.3	12.8	16.7	20.8	23.3	24.2	23.8	22.3	18.8	13.7	9.6
		MCDB	12.1	12.0	17.7	22.3	25.5	28.9	30.0	28.7	26.3	22.2	16.8	12.2
	10%	WB	5.6	6.1	10.5	15.1	19.6	22.4	23.4	23.0	21.2	16.9	11.5	6.8
		MCDB	8.2	9.7	15.0	20.3	24.1	27.4	28.8	27.5	24.8	20.4	14.6	9.4
Mean Daily Temperature Range	5% DB	MDBR	8.8	9.7	10.9	12.3	11.8	11.5	11.4	11.3	11.6	11.7	10.7	8.9
		MCDBR	13.3	15.0	15.7	16.6	15.0	13.5	13.5	13.0	14.6	15.1	14.8	12.9
		MCWBR	9.9	10.3	9.7	8.6	7.3	6.0	4.9	5.3	6.2	8.3	9.8	9.4
	5% WB	MCDBR	12.7	13.3	14.3	13.9	12.8	11.9	11.4	11.0	11.1	12.3	12.1	11.4
		MCWBR	10.0	9.7	9.7	8.1	6.7	5.7	4.8	5.0	5.5	7.4	9.2	9.2
Clear-Sky Solar Irradiance	taub	0.307	0.324	0.357	0.390	0.428	0.495	0.514	0.511	0.428	0.368	0.323	0.309	
	taud	2.450	2.379	2.364	2.301	2.219	2.065	2.031	2.022	2.233	2.401	2.490	2.477	
	Ebn at Noon	864	895	898	885	854	795	775	765	812	834	839	834	
	Edh at Noon	80	99	112	127	141	164	169	166	126	96	77	73	
All-Sky Solar Radiation	RadAvg	1.92	2.79	3.72	4.76	5.31	5.89	5.79	5.18	4.24	3.12	2.23	1.68	
	RadStd	0.17	0.34	0.21	0.39	0.54	0.45	0.33	0.27	0.55	0.36	0.20	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	-0.40	N/A	N/A	N/A	N/A
Regional (61 neighbors)		+0.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+120	+63

Nomenclature: See separate page